

GReinSS

Generative Modeling of Discrete Latent Structures via Dynamic Policy Gradients

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NCI Spring School on Algorithmic Cancer Biology — Tutorial

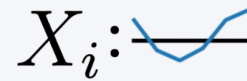
Ivanovic et al., ICML 2026

A recurring statistical inference problem in computational biology

States $S_{1:N} \sim \Pr^*(\mathcal{S})$



Measurements $X_{1:N}$
generated from $S_{1:N}$

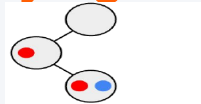


$\mathcal{S}$ is typically **large and combinatorial** — graphs, strings, sets, ...



Rather than directly observing the latent **state** S we care about, we observe some indirect **measurement** X generated from it.

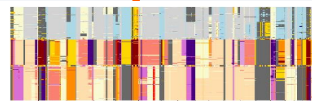
Phylogenies



State: tumor evolution tree
Measurement: DNA-seq

[Ivanovic & El-Kebir, RECOMB/Genome Res. 2023]

CNA profiles



State: copy-number profile
Measurement: read depth + BAF

[Ivanovic & El-Kebir, Genome Biol. 2025]

RNA isoforms

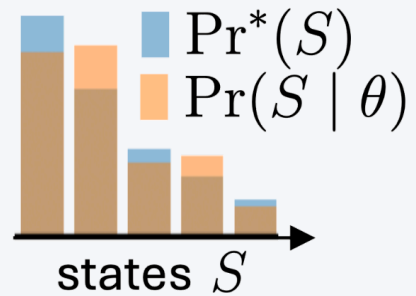


State: spliced transcript
Measurement: aligned short reads

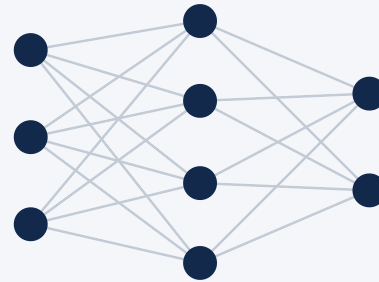
[Ivanovic et al., ICML 2026]

A learning and an inference problem

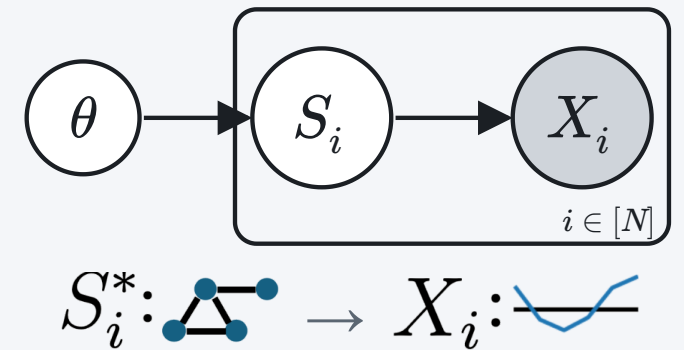
Approximate $\Pr^*(S)$ as
 $\Pr(S \mid \theta)$



Parameters θ : a linear map
or neural network



Generative model with
given $\Pr(X \mid S)$



Problem 1 (Learning). Given (i) $X_{1:N}$ and (ii) $\Pr(X \mid S)$, find θ maximizing
 $\Pr(X_{1:N} \mid \theta) = \prod_i \Pr(X_i \mid \theta)$, where $\Pr(X_i \mid \theta) = \sum_S \Pr(X_i \mid S) \Pr(S \mid \theta)$

Problem 2 (Inference). Given (i) $X_{1:N}$, (ii) $\Pr(X \mid S)$, and (iii) θ , find $\hat{S}_{1:N}$, where
 $\hat{S}_i = \arg \max_S \Pr(X_i \mid S) \Pr(S \mid \theta)$

Why the usual tools struggle

Approach	Problem
Expectation–Maximization	E-step expectation over $\mathcal{S}$ is intractable when $\mathcal{S}$ is combinatorial
Variational inference	maximizes an <i>ELBO</i> bound, not the likelihood — needs a tractable posterior family over combinatorial $\mathcal{S}$
Variational autoencoders	learn <i>artificial</i> continuous latents — not the mechanistic $\mathcal{S}$ you want
Local search ($\arg \max_S \Pr(X_i \mid S)$)	ignores the shared model $\Pr(S \mid \theta)$ across observations
Naive policy gradient	collapses to the single highest-reward state
GFlowNets	need known terminal rewards , not a likelihood to maximize

Gap: none of these directly maximize $\Pr(X_{1:N} \mid \theta)$ over a *discrete, combinatorial* state space.

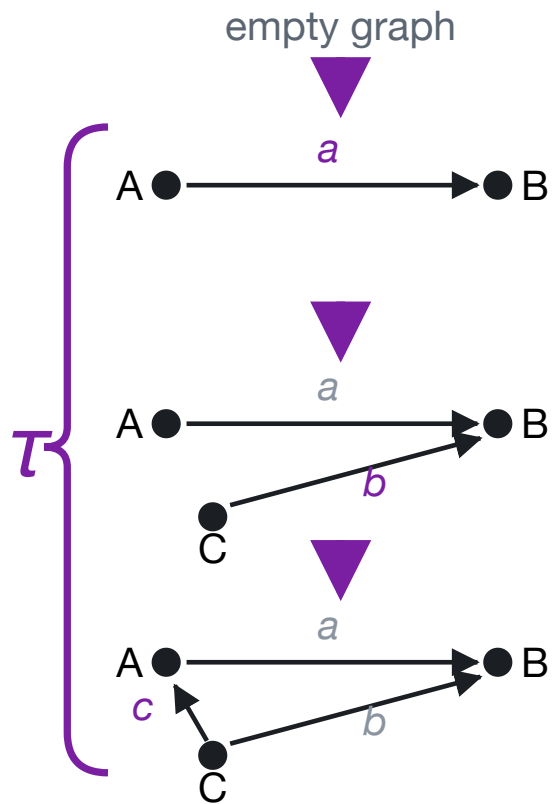
The GReinSS idea

Build the discrete state step by step with a policy, and reward trajectories by their share of each observation's likelihood.

Generative Reinforcement Learning of Structured States

Generate states as trajectories

A **policy** with parameters θ builds a state by a sequence of actions — a **trajectory** τ ending in a terminal state $S(\tau) \in \mathcal{S}$.



τ = the sequence of actions; $S(\tau)$ is its terminal state

- add an **edge** $\rightarrow$ a graph
- add an **element** $\rightarrow$ a set
- add a **junction** $\rightarrow$ an isoform

The policy is a **neural network** whose weights are the parameters θ we learn.

Sequential construction makes *combinatorial* spaces tractable: no summation over $\mathcal{S}$, just **sampling** trajectories.

$$\Pr(X \mid \theta) = \mathbb{E}_{\tau \sim \Pr(\tau \mid \theta)} [\Pr(X \mid S(\tau))]$$

The one equation that matters

Train with a policy gradient using the **dynamically rescaled reward** $r(\tau) = \sum_{i=1}^N \frac{\Pr(X_i \mid \tau)}{\Pr(X_i \mid \theta)}$.

$$\frac{d}{d\theta} \log \Pr(X_{1:N} \mid \theta)$$

what we optimize

=

$$\mathbb{E}_{\tau} \left[r(\tau) \frac{d}{d\theta} \log \Pr(\tau \mid \theta) \right]$$

how we optimize it

Theorem 1 (Unbiased policy gradient). *With the dynamically changing reward $r(\tau) = \sum_{i=1}^N \Pr(X_i \mid \tau) / \Pr(X_i \mid \theta)$, the policy gradient $\mathbb{E}_{\tau} \left[r(\tau) \frac{d}{d\theta} \log \Pr(\tau \mid \theta) \right]$ is an unbiased estimator of $\frac{d}{d\theta} \log \Pr(X_{1:N} \mid \theta)$, the gradient of the log-likelihood objective.*

So standard policy-gradient ascent with this reward = **maximum-likelihood** learning of $\Pr(S \mid \theta)$.

Why the denominator? (intuition)

The rescaling rewards a trajectory for its **proportional contribution** to $\Pr(X_i \mid \theta)$ — not its raw probability. This makes the policy **spread** over the states the data needs, instead of collapsing to one.

Toy: 3 states, 2 observations.

$$\Pr(X_1|S_1) = .5, \Pr(X_2|S_2) = .3, \Pr(X_2|S_3) = .2$$

Naive PG → chases the max reward (τ_1):

$$\Pr(\tau_1) = 1 \Rightarrow \Pr(X_2 \mid \theta) = 0 \\ \Rightarrow \Pr(X_{1:N} \mid \theta) = \mathbf{0}$$

✗

GReinSS → rewards balance out:

$$\Pr(\tau_1) = \Pr(\tau_2) = 0.5, \Pr(\tau_3) = 0 \\ \Rightarrow \Pr(X_{1:N} \mid \theta) = 0.0375 \checkmark \text{ (optimal)}$$

a	b	c	d
$\Pr(X \mid S)$	$S(\tau_i) = S_i$	$\uparrow \Pr(\tau_i \mid \theta) \Rightarrow \downarrow r(\tau_i)$	$\Pr(\tau_i \mid \theta)$
$S_1 \quad S_2 \quad S_3$	$S(\tau_1) = S_1$	$\uparrow \Pr(\tau_1 \mid \theta) \Rightarrow \downarrow r(\tau_1)$	$\Pr(\tau_1 \mid \theta) = 0.5$
$X_1 \quad 0.5 \quad 0 \quad 0$	$S(\tau_2) = S_2$	$\uparrow \Pr(\tau_2 \mid \theta) \Rightarrow \downarrow r(\tau_2)$	$\Pr(\tau_2 \mid \theta) = 0.5$
$X_2 \quad 0 \quad 0.3 \quad 0.2$	$S(\tau_3) = S_3$		$\Pr(\tau_3 \mid \theta) = 0$

The reward is *dynamic* — that's the whole trick

The denominator $\Pr(X_i \mid \theta)$ is **not known up front** and **changes every step** as θ updates — so we **re-estimate it by sampling** each iteration.

$$r(\tau) = \sum_{i=1}^N \frac{\Pr(X_i \mid \tau)}{\Pr(X_i \mid \theta)}$$

moves during training



$$\Pr(X_i \mid \theta) = \mathbb{E}_{\tau \sim \Pr(\tau \mid \theta)} [\Pr(X_i \mid \tau)]$$

estimated by sampling

Recomputed after every update, the moving denominator continuously **rebalances** which states the policy must cover. This is the "*dynamic*" in **dynamic policy gradients**.

The training loop

Repeat until convergence:

1. **Sample** trajectories $\tau \sim \text{Pr}(\tau \mid \theta)$
2. **Score** each with $\text{Pr}(X_i \mid \tau)$
3. **Rewards** $r(\tau) = \sum_i \text{Pr}(X_i \mid \tau) / \text{Pr}(X_i \mid \theta)$
4. **Policy-gradient step** on θ
5. **Recompute** $\text{Pr}(X_i \mid \theta) \rightarrow$ new rewards

You provide only two things:

- a way to **generate** S (grow it action-by-action)
- the likelihood $\text{Pr}(X \mid S)$

Everything else is generic.

Mini-batching over observations keeps it unbiased and scalable (Thm.).

Off-policy learning (when on-policy is too slow)

If sampling $\Pr(\tau \mid \theta)$ rarely produces states that explain any X_i , learning stalls.

Sample where the data says to look — bias toward the Bayes posterior:

Theorem 2 (Optimal off-policy proposal). *The unbiased, variance-minimizing sampling proposal is*

$$q(\tau \mid X_{1:N}, \theta) = \frac{1}{N} \sum_{i=1}^N \Pr(\tau \mid X_i, \theta), \quad \Pr(\tau \mid X_i, \theta) = \frac{\Pr(X_i \mid \tau) \Pr(\tau \mid \theta)}{\Pr(X_i \mid \theta)}.$$

Importance sampling keeps the gradient correct. *In cancer apps, a cheap heuristic (e.g. CNNaive in CNRein) seeds plausible states.*

The scoreboard — who actually maximizes the likelihood?

Method	Paradigm	Max. $\prod_i \Pr(X_i \theta)$?
GReinSS	RL	✓ Yes
Naive policy gradient	RL	✗ No
GFlowNets	RL	✗ No
VAE / autoregressive / diffusion + EM	EM	≈ approx.
Local search	per-observation	✗ No

One green **Yes**. The RL cousins optimize the *wrong* target; EM methods only *approximate* it; local search ignores the **shared** model entirely.

→ NOTEBOOK · Demo 1: Set reconstruction

Recover binary **sets** from noisy real-valued measurements. *Trains live in ~10 s.*

Problem. $S_i^* \subseteq \mathcal{U}$, observe

$X_{i,j} \sim \mathcal{N}(1, \sigma^2)$ if $j \in S_i^*$, else $\mathcal{N}(0, \sigma^2)$.

You'll write:

```
def log_pr_x_given_g(state, obs):  
    return -0.5*np.sum((obs-state)**2)/sigma**2
```

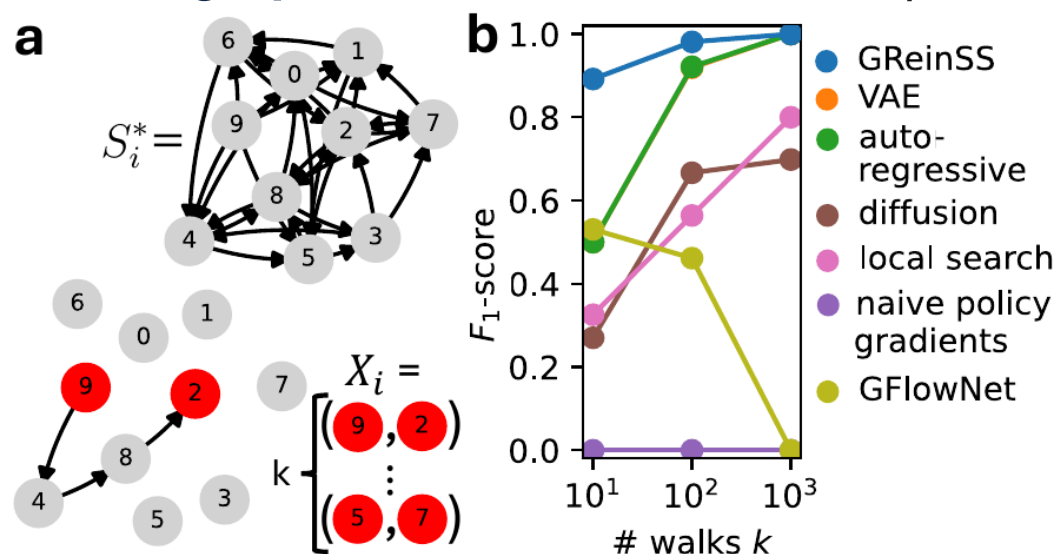
...then `simpleTrainModel(...)`.

Watch for:

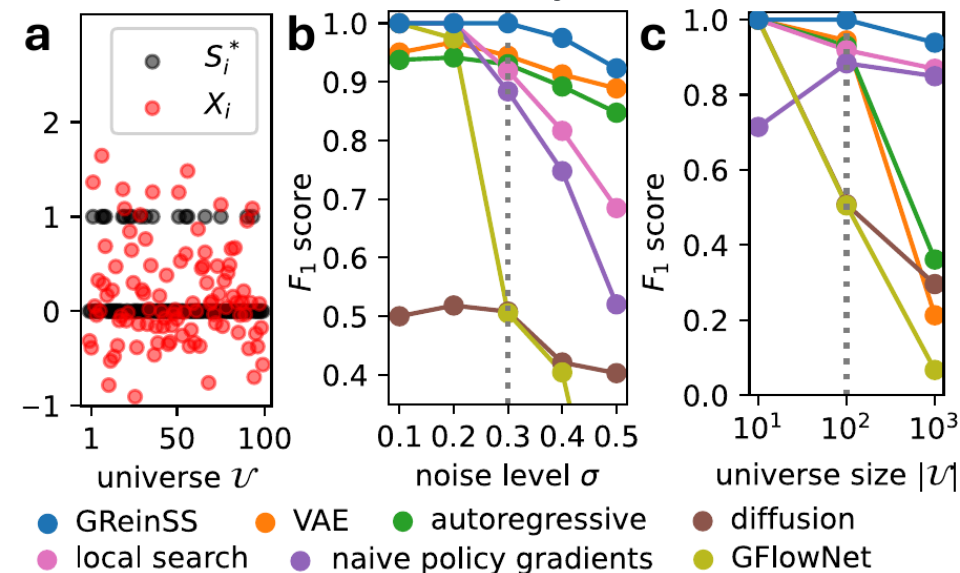
- median $\log \Pr(X_i \mid \theta)$ climbing to ~ 0
- recovered sets vs. thresholding the noise
- where the **shared model fixes** noisy bits

Results — simulations

Latent graphs from random-walk endpoints



Latent sets from noisy measurements



$k = 10$ walks: GReinSS $F_1 = \mathbf{0.891}$; all baselines < 0.55

$|\mathcal{U}| = 1000$: GReinSS $F_1 = \mathbf{0.938}$; GEM baselines < 0.4

Dynamic rewards are a **small** code change over naive PG — but decisive. Naive PG here predicts the **empty graph** ($F_1 = 0$).

→ NOTEBOOK · Demo 2: Graph inference (pre-trained)

Latent **directed graphs** from start/end points of k absorbing random walks.

State = 90 possible directed edges (10 nodes).

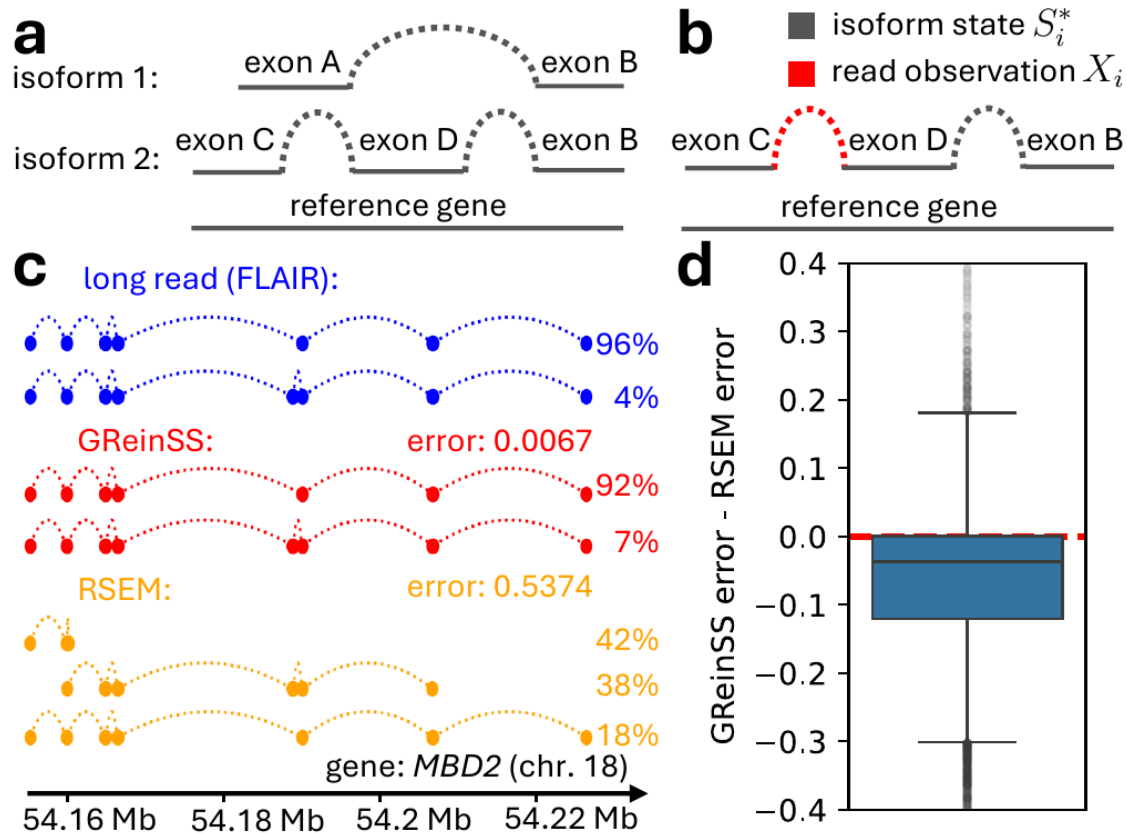
$\Pr(X \mid S)$ from the shifted-Laplacian $(L + I)^{-1}$ random-walk model.

We **load a pre-trained policy**, run inference, and score edge-recovery F_1 against ground-truth graphs.

Watch for:

- training likelihood curve (pre-computed)
- a reconstructed graph $\hat{S}_i$ vs. true S_i^*
- per-graph F_1 distribution

Application — RNA isoforms beat RSEM



State S = an isoform (chosen exon junctions) + sample + read position.

$\Pr(X | S) = 1$ iff read position & sample match — trivial forward model.

On **GTEx** (61 samples w/ matched long reads, 14,390 genes):

- GReinSS **beats RSEM by ≥ 0.05** on **46.6%** of genes; RSEM beats GReinSS on only **9.4%**
- *MBD2* example: GReinSS error **0.007** vs RSEM **0.537**

GReinSS already powers two cancer methods

CloMu

Tumor **phylogenies** of SNVs

States: mutation trees

Observations: noisy DNA-seq trees

Ivanovic & El-Kebir, RECOMB / Genome Res. 2023

CNRein

Copy-number evolution in single cells

States: sets of CNA events per cell

Observations: single-cell DNA-seq

Ivanovic & El-Kebir, Genome Biol. 2025

This paper **generalizes** the shared technique behind both into one framework for *any* discrete latent structure — with theory, off-policy theory, and baselines.

When should *you* reach for GReinSS?

Good fit ✓

- latent state is **discrete** / **combinatorial**
- you can **generate** it incrementally
- you know (or can compute) $\Pr(X \mid S)$
- observations are **indirect** & shared structure exists

Reach elsewhere ✗

- continuous latents → VAE / diffusion
- tractable exact E-step → classic EM
- rewards truly fixed & known → standard RL / GFlowNet

Recipe: ① write a generator for S · ② write $\Pr(X \mid S)$ · ③ `simpleTrainModel` · ④ `simpleInference` . *(optional)* add an off-policy proposal for hard instances.

Thank you — let's build

Code + notebook: `code/README.md`, `tutorial/GReinSS_demo.ipynb`

Recipe: generator for S + likelihood $\Pr(X \mid S) \rightarrow \text{train} \rightarrow \text{infer}$

Questions? · melkebir@illinois.edu